

Report: Certificate Services & Secure Authentication

1. Introduction

This set of tasks focused on implementing certificate-based authentication in an Active Directory environment using AD Certificate Services (AD CS). The process involved setting up an Enterprise Root Certificate Authority, issuing custom templates for users and computers, and enabling automated certificate enrollment via Group Policy. These steps enhance authentication security, enable encrypted communications, and form the foundation for secure services such as IPsec, EFS, and smart card logon. All configurations were carried out on the domain controller and verified through successful issuance of certificates to domain-joined clients and users.

2. Objective

The objective was to deploy a certificate infrastructure within the Active Directory domain. This included installing AD CS and configuring it as an Enterprise Root CA, creating and publishing customized certificate templates for user and computer authentication, and setting up auto-enrollment using GPOs. By automating certificate distribution, the goal was to improve authentication reliability and scalability while eliminating the need for manual certificate requests.

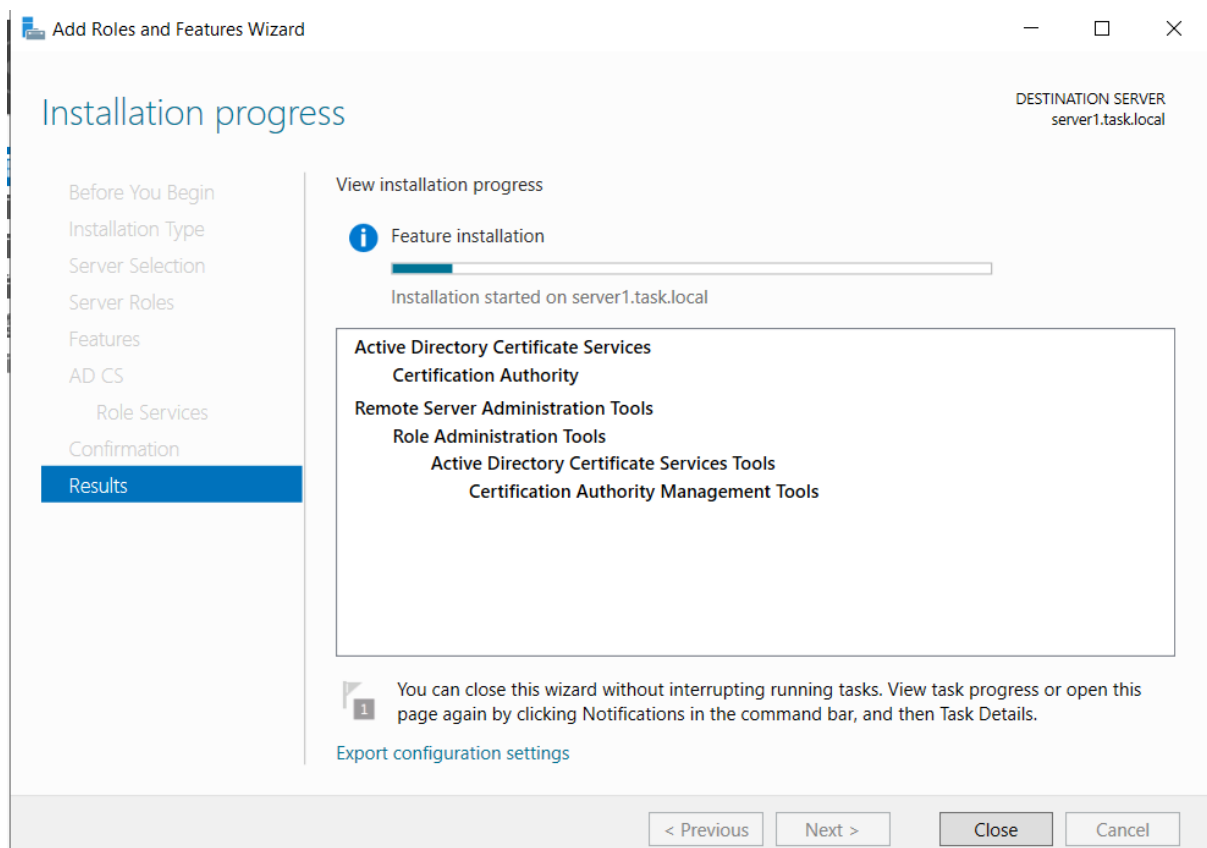
3. Tools used

- Server Manager
- AD CS (Certification Authority MMC)
- Certificate Templates Console (certtmpl.msc)
- Group Policy Management Console (gpmc.msc)
- MMC Snap-in for Certificates
- PowerShell (certutil, gpupdate)
- Active Directory Users and Computers (ADUC)

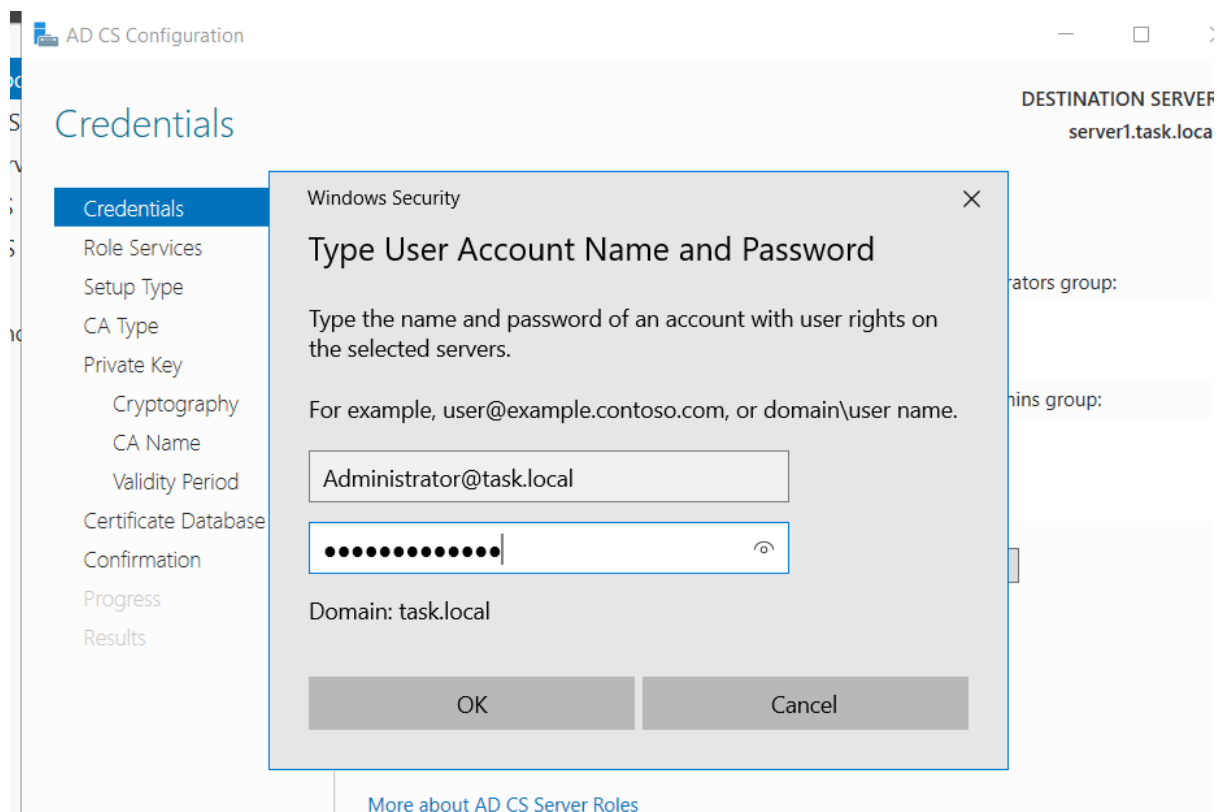
4. Methodology

4.1 Install AD Certificate Services

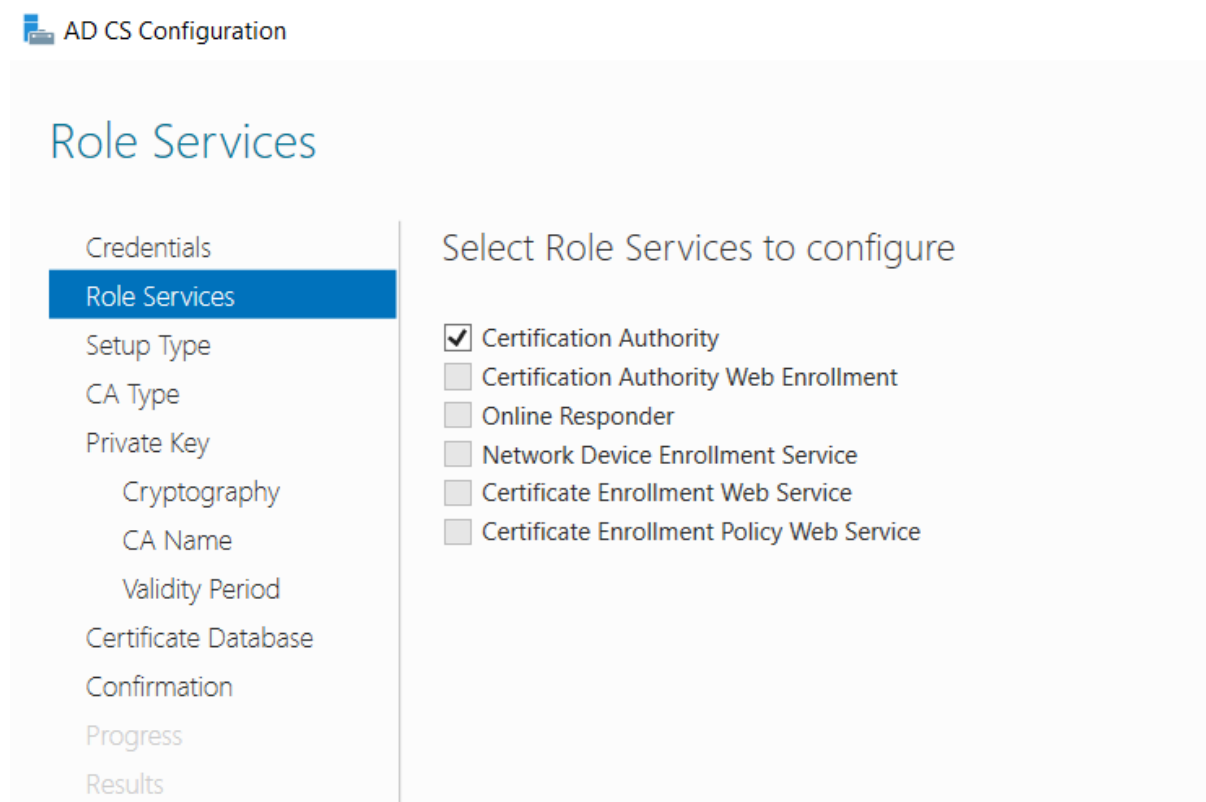
AD CS role and the Certification Authority service were installed on the domain controller (Server 2022). The CA was configured as an Enterprise Root CA using SHA256 and RSA 2048-bit encryption. The Certification Authority was verified in Server Manager and certsrv.msc.



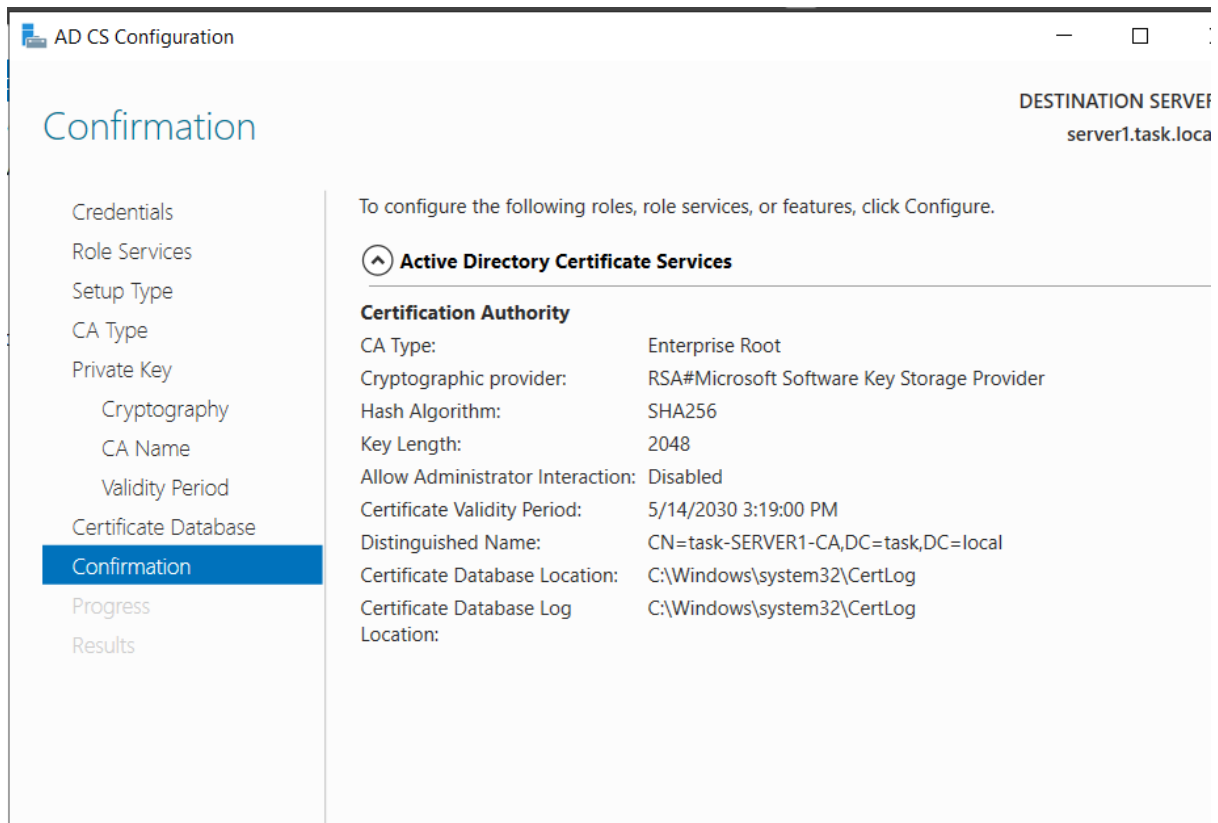
(Add active directory certificate services from server manager)



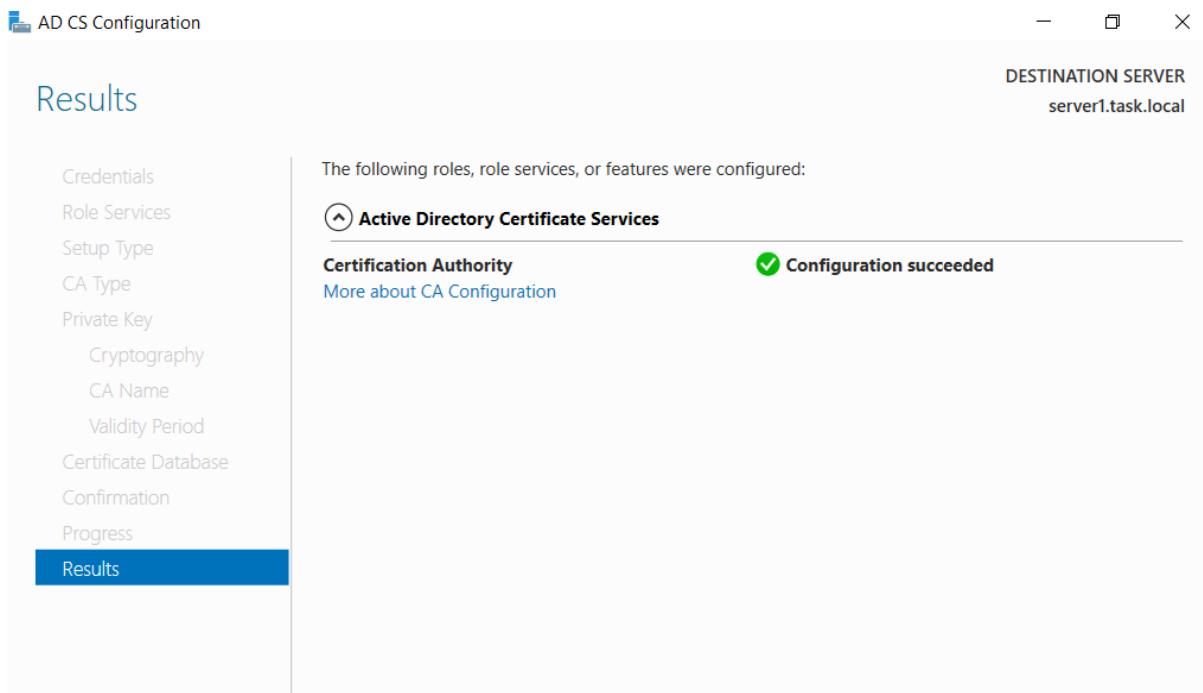
(Give credentials of your PC1 admin account)



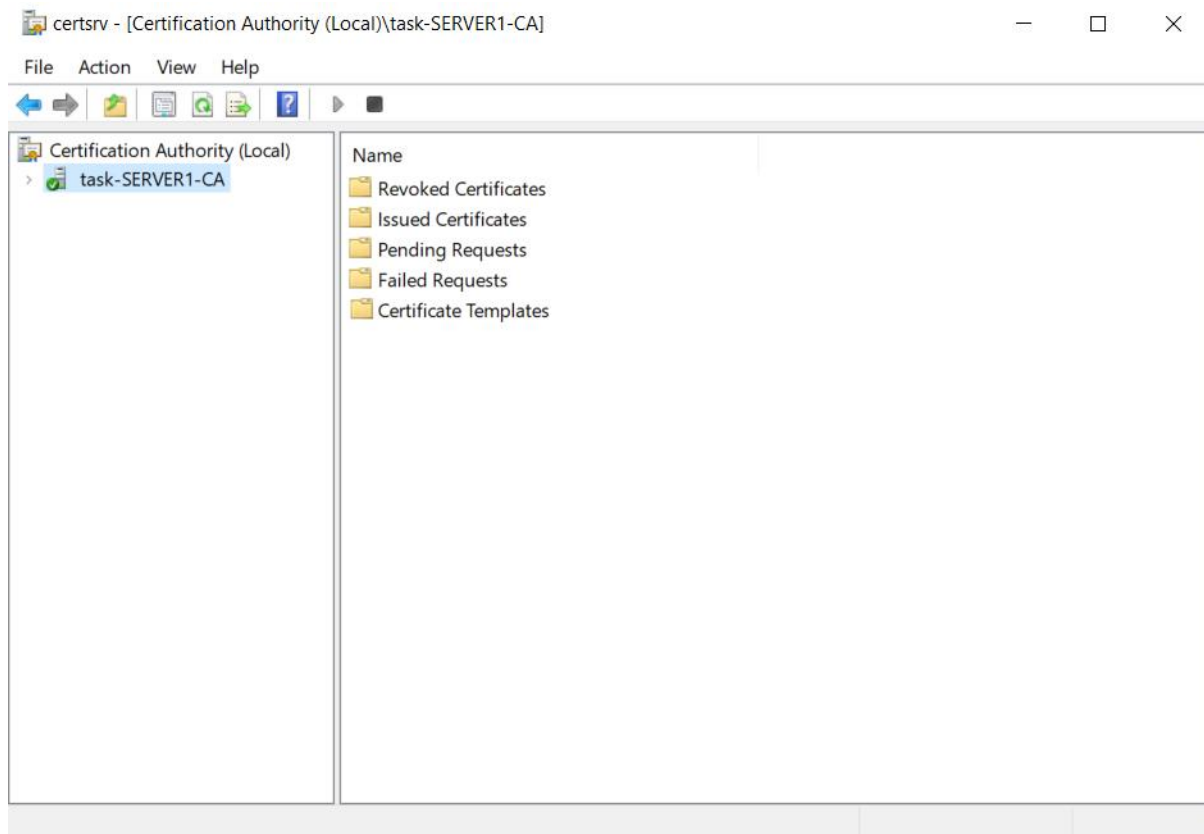
(Define role services and leave other configs default)



(Confirmation for all the setting's done)



(Configure all settings)



(Done successfully)

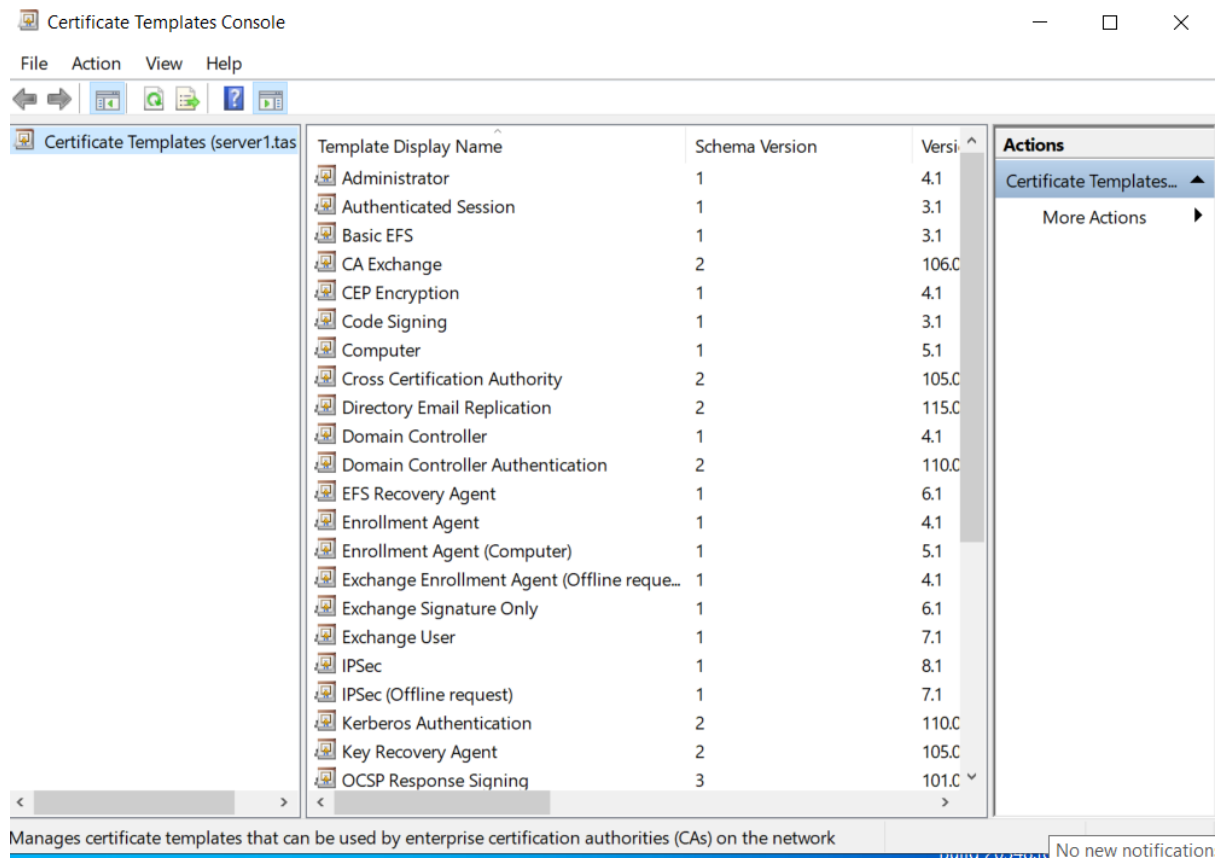
4.2 Issue Certificates for Domain Authentication

Custom certificate templates were created for both users and computers:

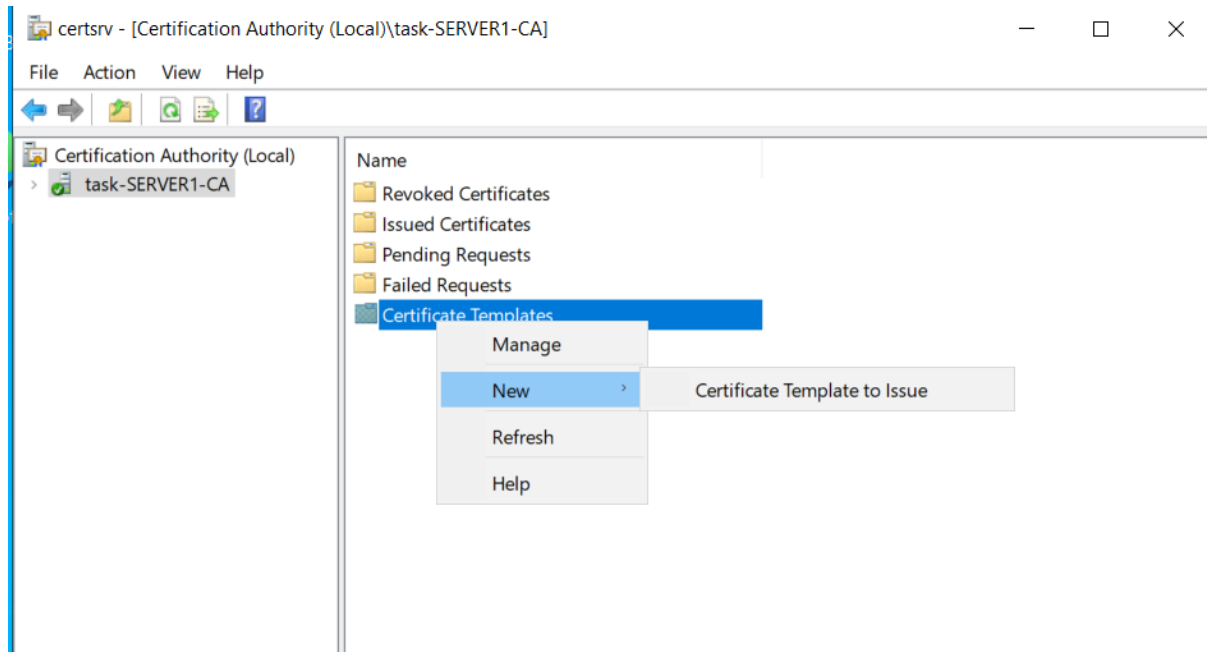
Custom User Auth

Custom Computer Auth

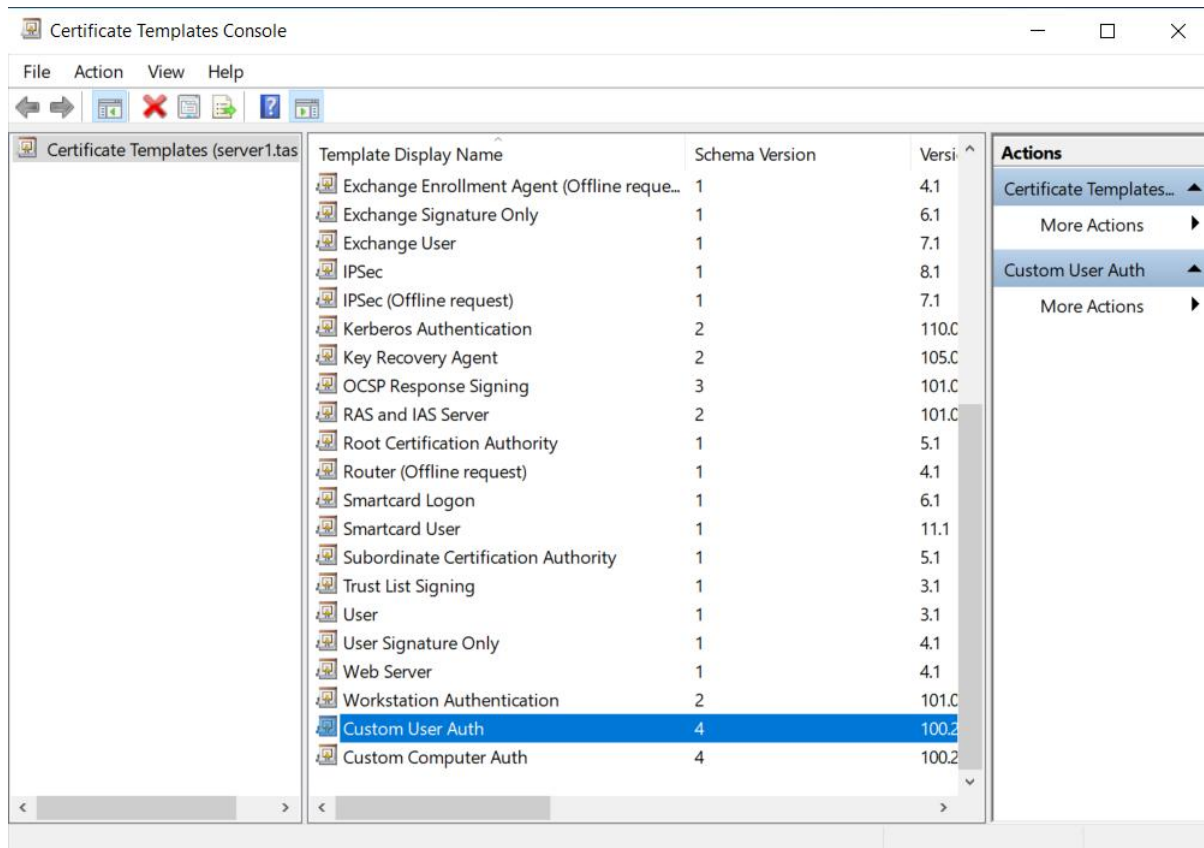
Templates were based on Windows Server 2016 compatibility, with auto-enrollment permissions assigned to Authenticated Users and Domain Computers. These templates were issued via the CA MMC console and confirmed under the Certificate Templates list.



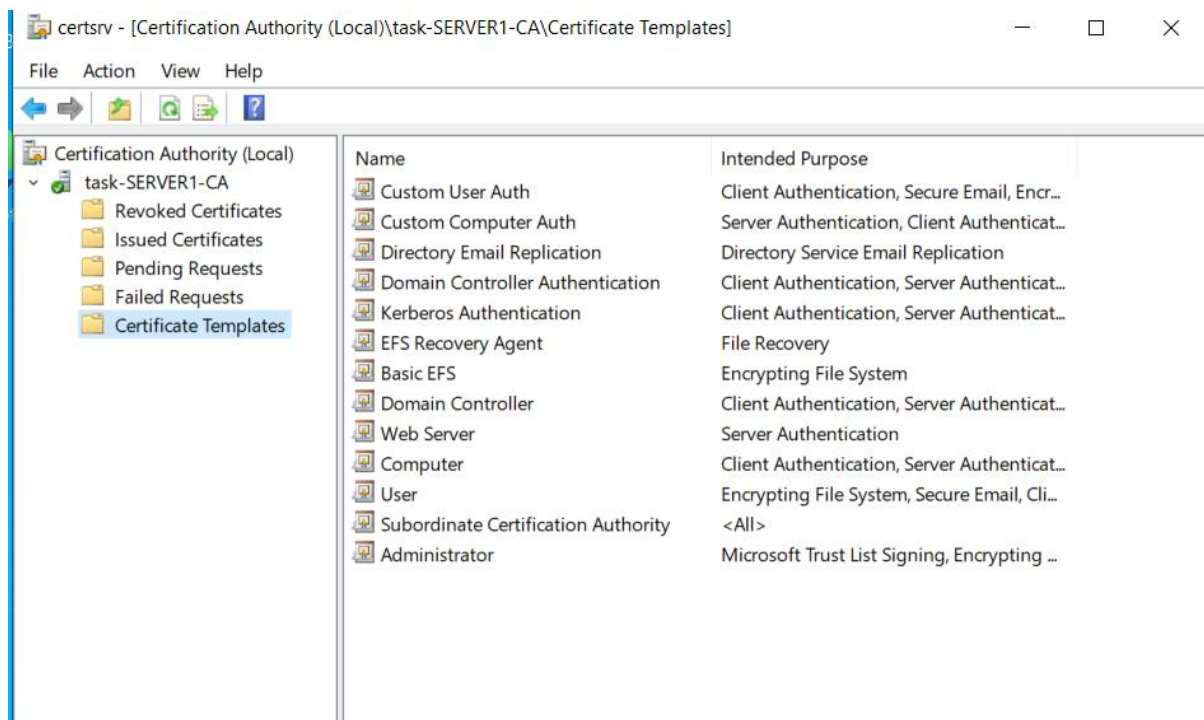
(open the Certificate Templates Console and here we give permission to enrol)



(Now open CA and add new template to certificate template)



(Created custom template for user and computer)



(Enrolled both certificates)

4.3 Configure Auto-Enrollment via GPO

A GPO was applied at the domain level to enable:

User Configuration → Certificate Services Client - Auto-Enrollment

Computer Configuration → Certificate Services Client - Auto-Enrollment

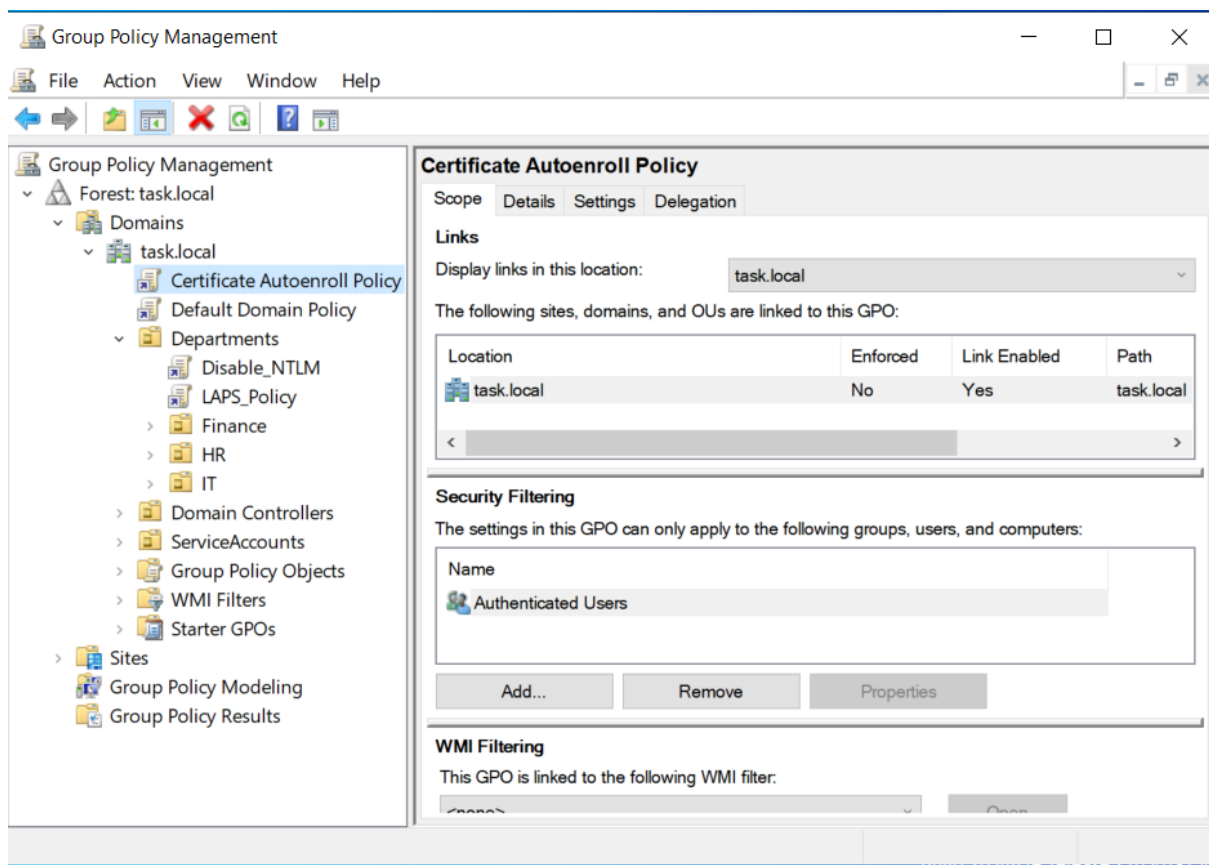
Both were set to "Enabled" with "Renew expired certificates" and "Update certificates" options checked. Group policy was updated on the client (pc1) using gpupdate /force.

Certificates were automatically issued to:

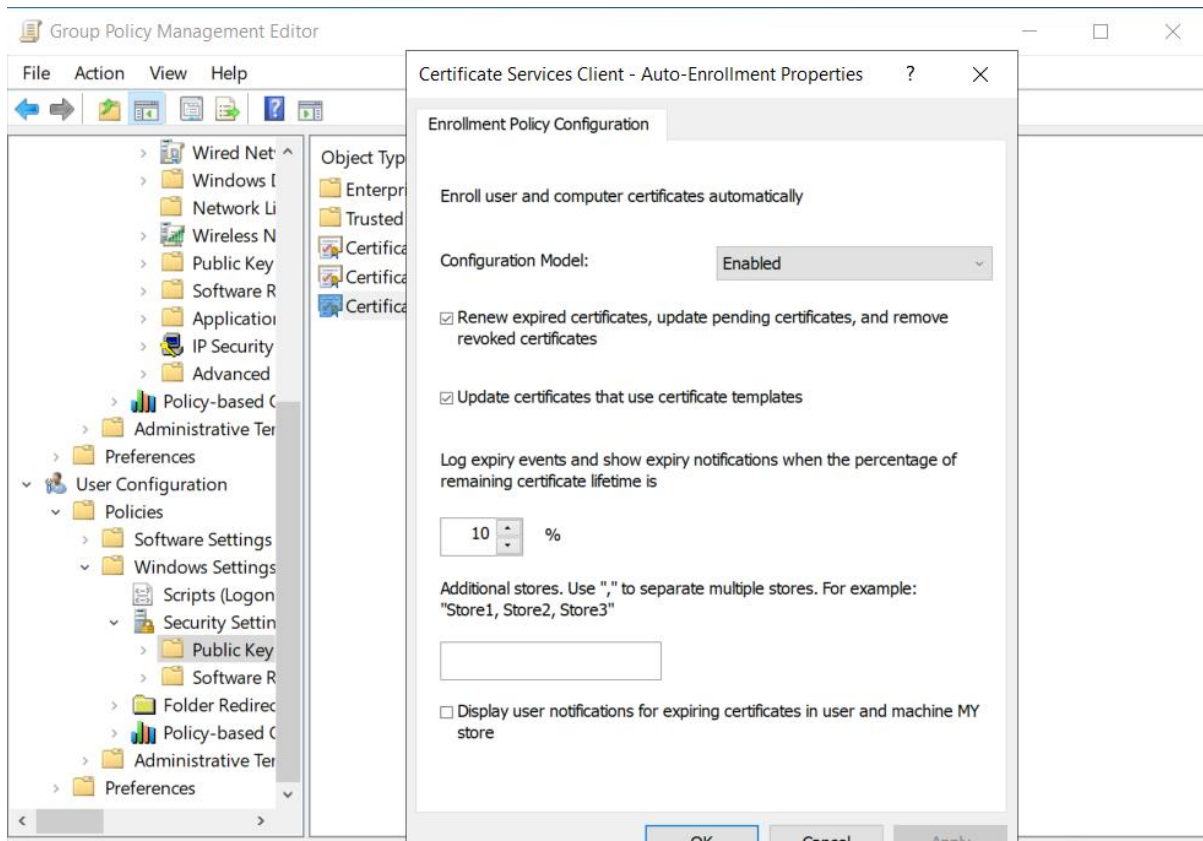
Aarav.Finance (user) → Custom User Auth

pc1.task.local (computer) → Custom Computer Auth

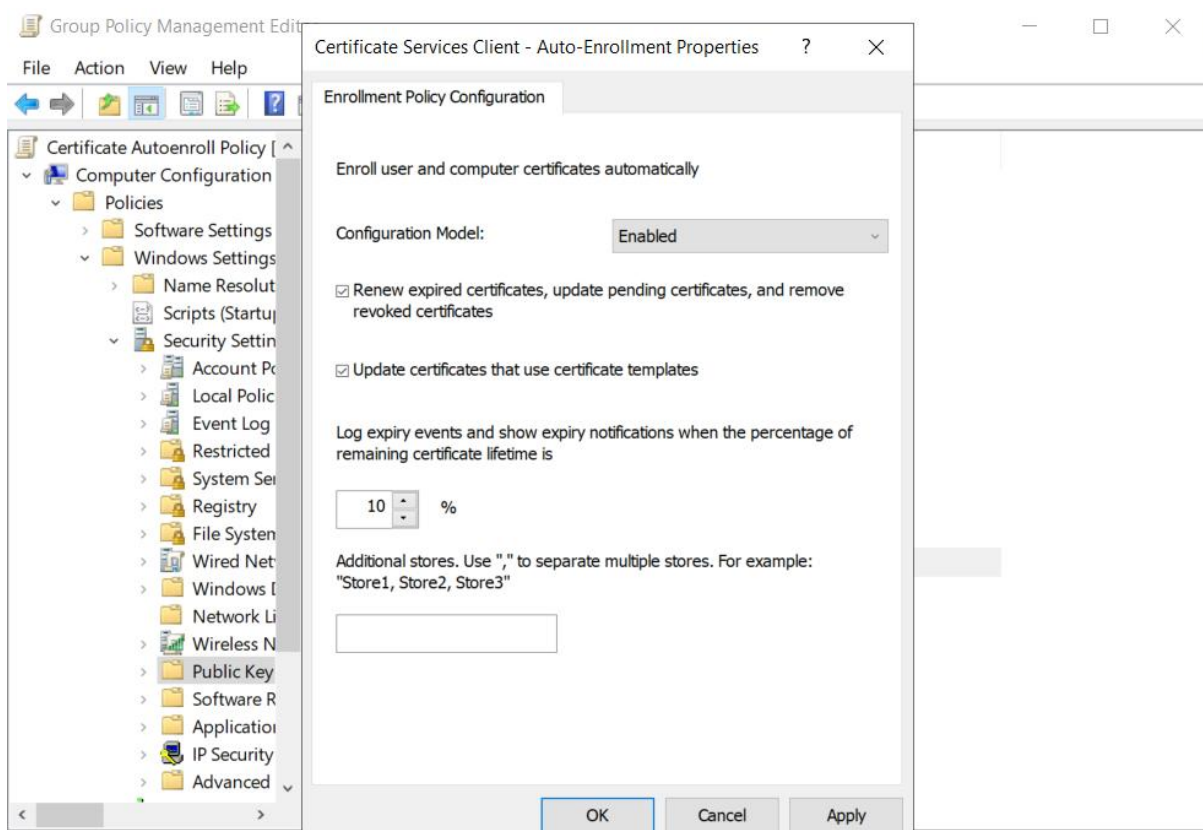
Enrollment was confirmed via certmgr.msc and mmc snap-ins.



(Create a new GPO for task.local)



(Edit policy – for user certificate)



(Edit policy – for computer certificate)

Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.19044.5854]
(c) Microsoft Corporation. All rights reserved.

C:\Users\administrator>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.

C:\Users\administrator>
```

(Updated policy)

```
C:\Users\administrator>gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on 5/14/2025 at 4:36:15 PM

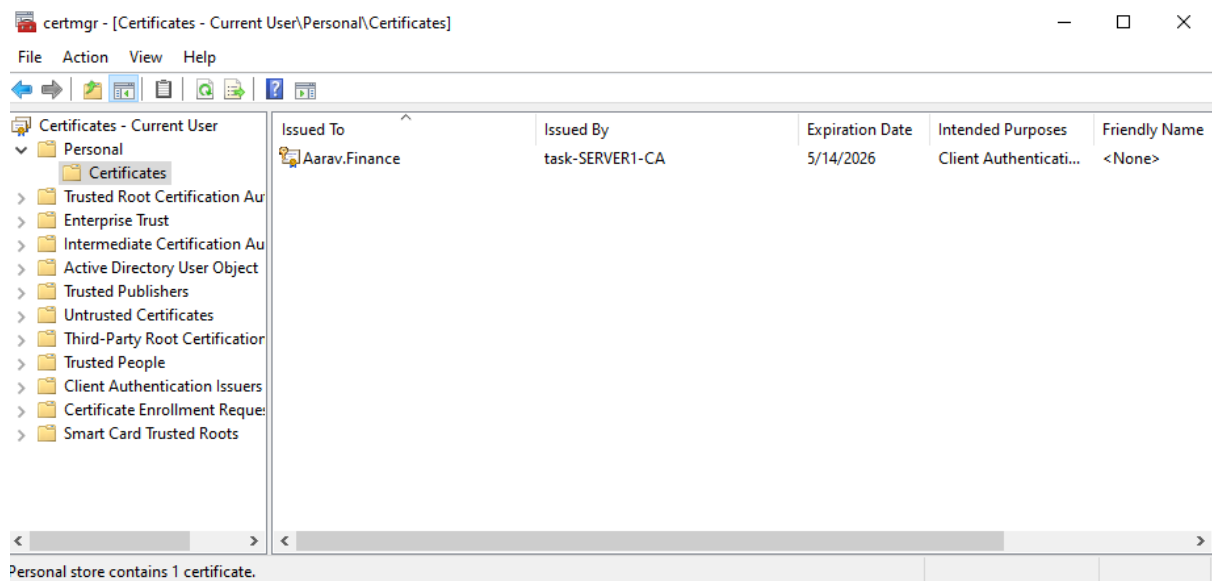
RSOP data for TASK\Administrator on PC1 : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:            10.0.19044
Site Name:             Default-First-Site-Name
Roaming Profile:       N/A
Local Profile:         C:\Users\administrator
Connected over a slow link?: No

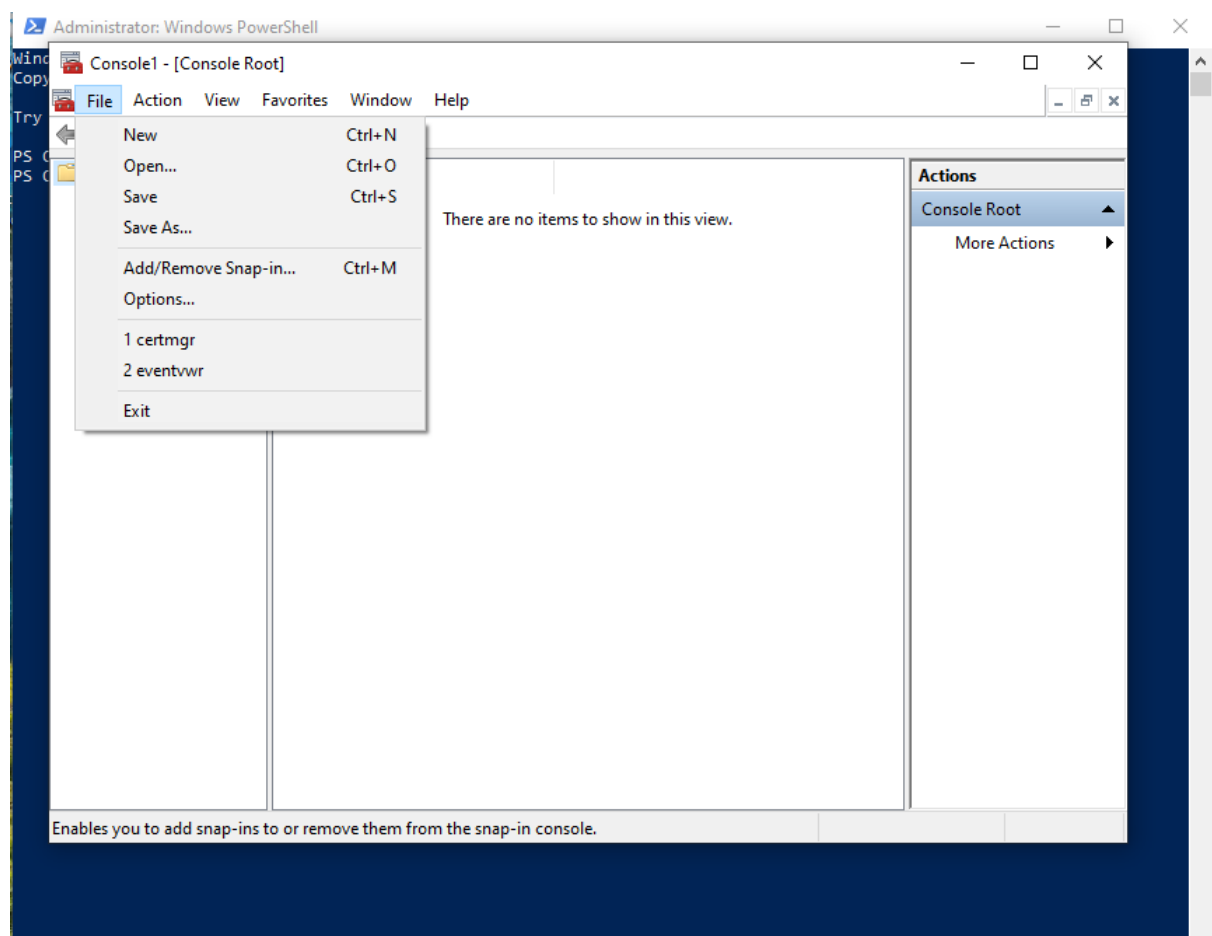
COMPUTER SETTINGS
-----
CN=PC1,OU=Finance,OU=Departments,DC=task,DC=local
Last time Group Policy was applied: 5/14/2025 at 4:35:22 PM
Group Policy was applied from:    server1.task.local
Group Policy slow link threshold: 500 kbps
Domain Name:                     TASK
Domain Type:                    Windows 2008 or later

Applied Group Policy Objects
-----
  Disable_USB_Finance
  Logon_Audit_Policy
  Object_Access_Audit
  ScreenLock_15Min
  Disable_SMBv1
  Logon_Banner_GPO
  Disable_NTLM
  LAPS_Policy
  Default_Domain_Policy
  Certificate_Autoenroll_Policy
```

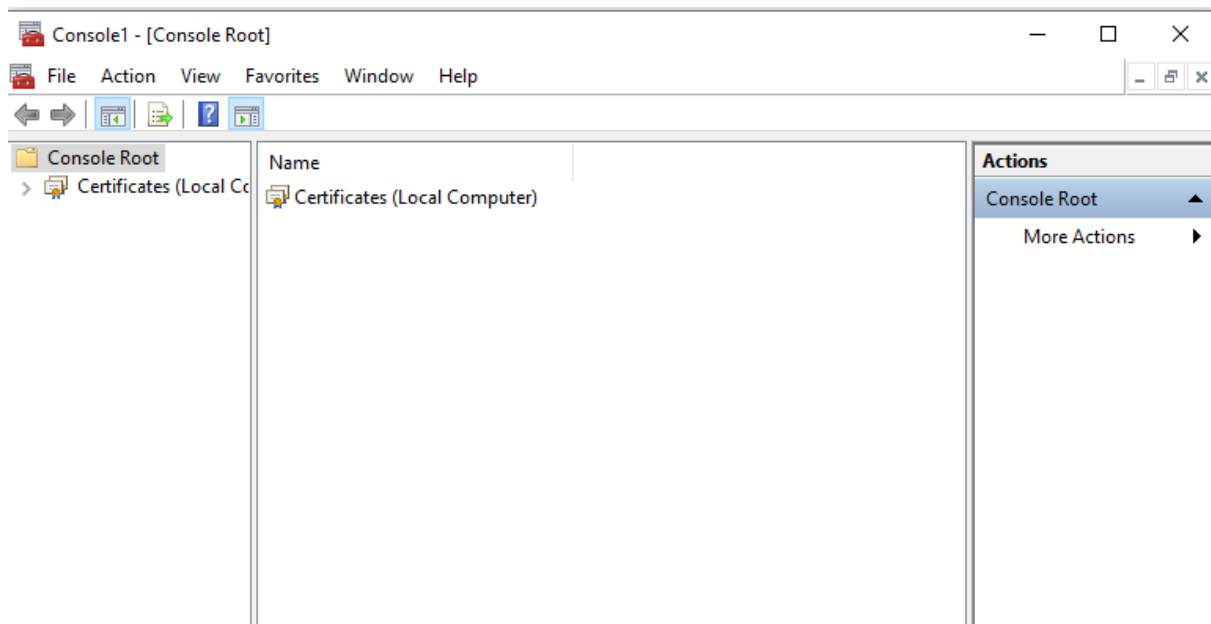
(Verified the policy is updated)



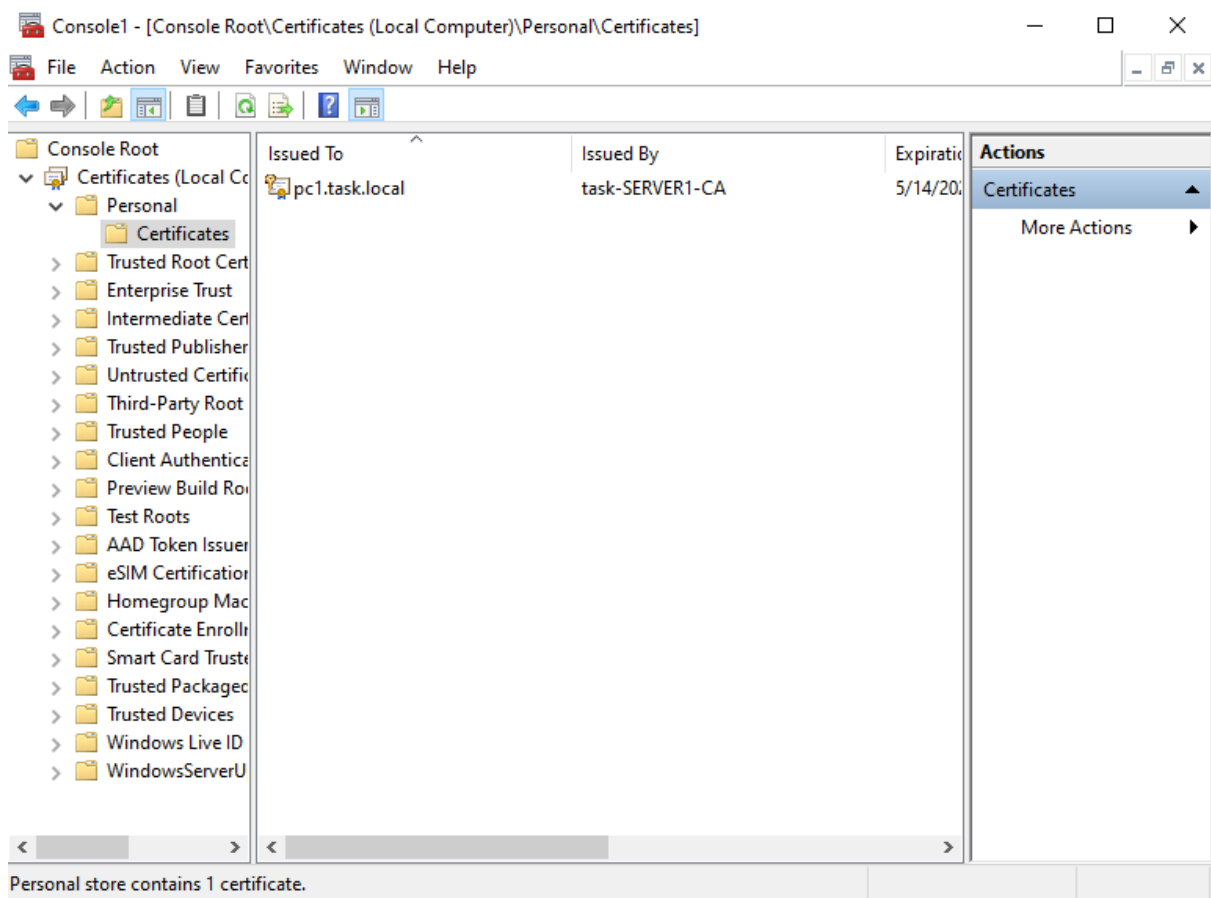
(We have our user certificate auto-enrolled here)



(open mmc console in user)



(Added a local computer cert.)



(Verified the certificate is here)

5. Recommendations

- Regularly back up the CA database and private key
- Enforce certificate expiration alerts and auto-renewal
- Limit certificate template permissions to only required security groups
- Periodically review issued certificates for misuse or anomalies
- Integrate with smart card or MFA systems for enhanced identity assurance
- Monitor CA logs and event IDs for enrollment activity

6. Conclusion

Certificate Services were successfully implemented in the AD environment with the setup of a secure and scalable PKI using an Enterprise Root CA. Customized templates were created and issued for domain users and machines, followed by a GPO-based configuration of auto-enrollment. The end result demonstrated a fully functional certificate lifecycle — from CA setup to issuance and renewal — enhancing both identity verification and secure service access across the domain.